



Centrum Wiskunde & Informatica

Uncertainty-aware surrogate modeling via conformal prediction

Dimitrios Loukrezis

Centrum Wiskunde & Informatica (CWI)

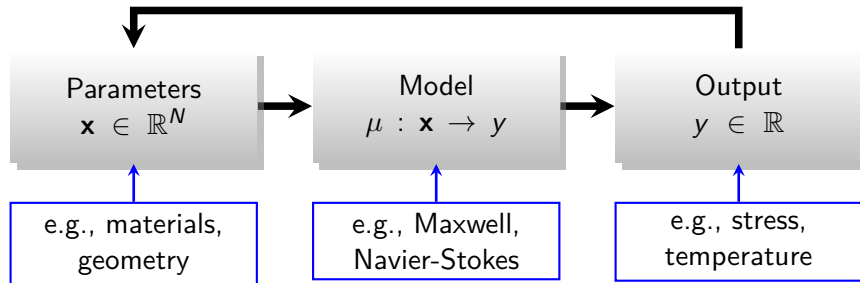
Workshop on *Industrial Applications of Numerical Analysis and Machine Learning*
RSP *Bridging Numerical Analysis and Scientific Machine Learning: Advances and Applications*

1. December 2025

- 1 Surrogate modeling (data-driven)
- 2 Conformalized surrogate models
- 3 Engineering application
- 4 Summary & outlook

- 1 Surrogate modeling (data-driven)
- 2 Conformalized surrogate models
- 3 Engineering application
- 4 Summary & outlook

Parametric problems and models



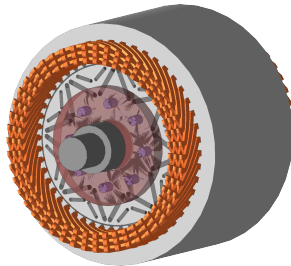
- Forward and inverse problems
- Deterministic and stochastic settings
- Estimation, optimization, uncertainty quantification, control, ...

Fidelity versus complexity trade-off

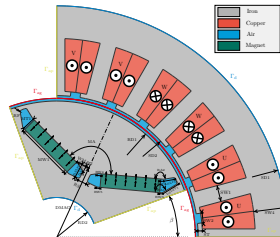
**Traction
motor**



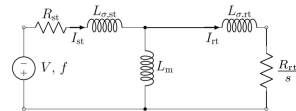
**3D model
full geometry**



**2D model
1/4 geometry**



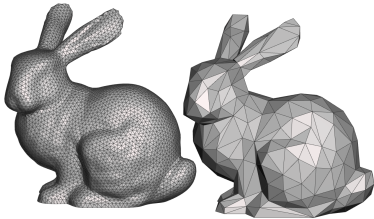
**Equivalent
circuit model**



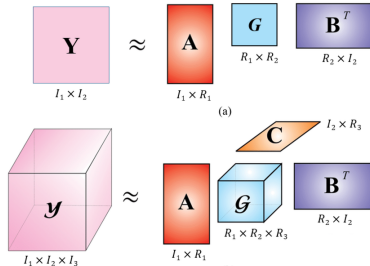
- Computational cost of high-fidelity models can be prohibitive for parameter studies
- **Surrogate modeling** sacrifices accuracy for computational efficiency

A **surrogate model** is an **approximate** mathematical model that simplifies a complex simulation, allowing for **faster evaluations**.

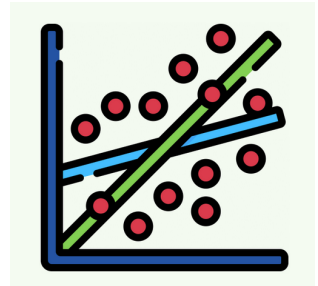
Model simplification



Model order reduction



Model learning



Surrogate model

$$\hat{\mu}(\mathbf{x}; \boldsymbol{\theta}) \approx \mu(\mathbf{x})$$

Find $\boldsymbol{\theta}$ that minimizes

$$\|\hat{\mu}(\mathbf{x}) - \mu(\mathbf{x})\|_{(\cdot)}$$

Training dataset

$$\mathcal{D}_{\text{train}} = \left\{ \mathbf{x}^{(m)}, y^{(m)} \right\}_{m=1}^M$$

Experimental design

$$\left\{ \mathbf{x}^{(m)} \right\}_{m=1}^M$$

Observations

$$\left\{ y^{(m)} = \mu(\mathbf{x}^{(m)}) \right\}_{m=1}^M$$

$$\boldsymbol{\theta} = \arg \min_{\hat{\boldsymbol{\theta}}} \left\{ \frac{1}{M} \sum_{m=1}^M \left(\hat{\mu}(\mathbf{x}^{(m)}; \hat{\boldsymbol{\theta}}) - y^{(m)} \right)^2 \right\}$$

Predictive uncertainty of regression models

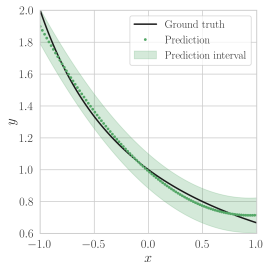
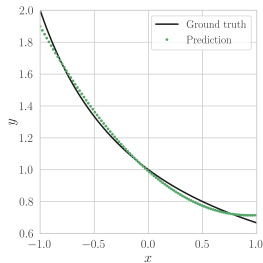
Regression model $\hat{\mu}(\mathbf{x}) \approx \mu(\mathbf{x})$ trained on $\mathcal{D}_{\text{train}} = \{\mathbf{x}^{(m)}, y^{(m)}\}_{m=1}^M$

Test point $(\mathbf{x}^*, y^*) \notin \mathcal{D}_{\text{train}} \longrightarrow$ Prediction $\hat{y}^* = \hat{\mu}(\mathbf{x}^*) \neq y^*$

Question: Can we complement the model's prediction with an interval that is **likely** to contain the true observation y^* ?

Formally: Given a **significance level** $s \in (0, 1)$, equivalently, a target **coverage level** $1 - s$, find an interval $\hat{\mathcal{C}}_{1-s}(\mathbf{x}^*)$ such that

$$\mathbb{P}\{y^* \in \hat{\mathcal{C}}_{1-s}(\mathbf{x}^*)\} \geq 1 - s.$$



- 1 Surrogate modeling (data-driven)
- 2 Conformalized surrogate models
- 3 Engineering application
- 4 Summary & outlook

Conformal prediction (for regression)

Non-conformity function $\alpha(\mathbf{x}, y; \hat{\mu})$ measures data conformity to fitted model

Calibration dataset $\mathcal{D}_{\text{cal}} = \{\mathbf{x}^{(j)}, y^{(j)}\}_{j=1}^J$

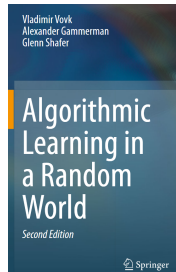
Non-conformity scores $\alpha_j = \alpha(\mathbf{x}^{(j)}, y^{(j)}; \hat{\mu})$

Here: **Absolute residuals** $\alpha_j = R_j = |y^{(j)} - \hat{\mu}(\mathbf{x}^{(j)})|$

$(1 - s)$ -**quantiles** of empirical non-conformity score distribution: $\hat{q}_{1-s} \{R_j\}$

$$\hat{C}_{1-s}(\mathbf{x}^*) = [\hat{\mu}(\mathbf{x}^*) - \hat{q}_{1-s} \{R_j\}, \hat{\mu}(\mathbf{x}^*) + \hat{q}_{1-s} \{R_j\}]$$

Assumption: Joint data distribution is invariant to permutation (**data exchangeability**)



Conformal prediction variants

Assume an available dataset $\mathcal{D} = \{\mathbf{x}^{(m)}, y^{(m)}\}_{m=1}^M$

Split CP: Split $\mathcal{D}$ into disjoint datasets $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{cal}}$

Compute $\hat{C}_{1-s}$ as before

- ✓ $(1-s)$ coverage guarantee; single model training
- ✗ Fewer training data $\rightarrow$ reduced model accuracy

Cross CP: Split $\mathcal{D}$ into Q disjoint datasets $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_Q$

Train Q models $\hat{\mu}_{\sim q}$ on the datasets $\mathcal{D}_{\sim q} = \mathcal{D} \setminus \mathcal{D}_q$

Non-conformity scores: **Cross-validation residuals** $R_m^{\text{cv}} = |y^{(m)} - \hat{\mu}_{\sim q(m)}(\mathbf{x}^{(m)})|$

- ✓ $(1-s)$ *empirical* coverage; full dataset used for training
- ✗ Q models trained; no coverage guarantee

Jackknife conformal prediction

! Extreme case of cross CP !

- Non-conformity scores: Absolute **leave-one-out (LOO) residuals**

$$R_m^{\text{LOO}} = \left| y^{(m)} - \hat{\mu}_{\sim_m}(\mathbf{x}^{(m)}) \right|$$

where $\hat{\mu}_{\sim_m}$ is trained with $\mathcal{D}_{\sim_m} = \mathcal{D} \setminus \{(\mathbf{x}^{(m)}, y^{(m)})\}$

$$\hat{C}_{1-s}^{\text{jack}}(\mathbf{x}^*) = \left[\hat{\mu}(\mathbf{x}^*) - \hat{q}_{1-s} \left\{ R_m^{\text{LOO}} \right\}, \hat{\mu}(\mathbf{x}^*) + \hat{q}_{1-s} \left\{ R_m^{\text{LOO}} \right\} \right]$$

- Repeated model training can be avoided for several regression models
 - E.g., Random forest, extreme learning machine, **polynomial chaos expansion (PCE)**

Polynomial chaos expansion (regression-based)

$$\hat{\mu}(\mathbf{x}) = \sum_{k=1}^K c_k \Psi_k(\mathbf{x})$$

- Polynomial basis $\{\Psi_k\}_{k=1}^K$; orthogonal w.r.t. to input probability measure
- Coefficients $\mathbf{c} = (c_1, \dots, c_K)^\top$
- Training data $\{\mathbf{x}^{(m)}, y^{(m)}\}_{m=1}^M$; output vector $\mathbf{y} = (y^{(1)}, \dots, y^{(M)})^\top$
- Design matrix $\mathbf{D} \in \mathbb{R}^{M \times K}$, $d_{mk} = \Psi_k(\mathbf{x}^{(m)})$

$$\mathbf{c} = \arg \min_{\hat{\mathbf{c}} \in \mathbb{R}^K} \left\{ \frac{1}{M} \|\mathbf{y} - \mathbf{D}\hat{\mathbf{c}}\|_2^2 \right\} = (\mathbf{D}\mathbf{D}^\top)^{-1} \mathbf{D}^\top \mathbf{y}$$

Conformalized polynomial chaos regression with the jackknife



D. Loukrezis and D.G. Giovanis, Conformalized polynomial chaos expansion for uncertainty-aware surrogate modeling, *arXiv preprint arXiv:2510.22375* (2025)

- Closed-form expression for R_m^{LOO} :

$$R_m^{\text{LOO}} = \left| \frac{y^{(m)} - \hat{\mu}(\mathbf{x}^{(m)})}{1 - h_{mm}} \right|$$

where h_{mm} are the diagonal terms of $\mathbf{H} = \mathbf{D}(\mathbf{D}\mathbf{D}^\top)^{-1}\mathbf{D}^\top$.

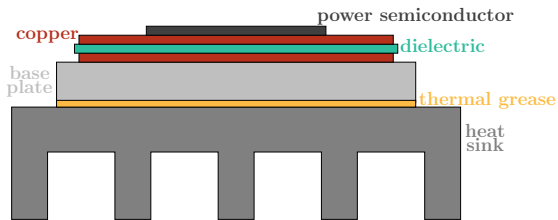
- Closed-form expression also for the [jackknife+](#) extension ([1 – 2s coverage guarantee](#))



R.F. Barber et al., Predictive inference with the jackknife+, *The Annals of Statistics* (2021)

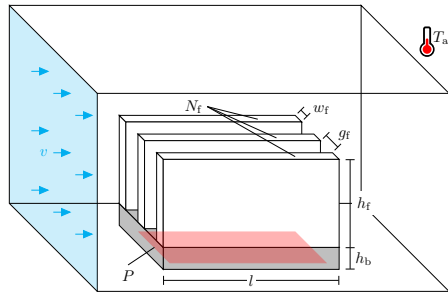
- 1 Surrogate modeling (data-driven)
- 2 Conformalized surrogate models
- 3 Engineering application**
- 4 Summary & outlook

Power electronics cooling



Power electronics module (IGBT)

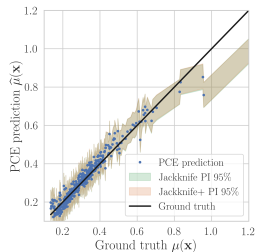
- Quantity of interest: $R_{th} = \frac{T_{hs} - T_a}{P}$
- 9 geometric / operational parameters
- Simulation time ≈ 35 minutes



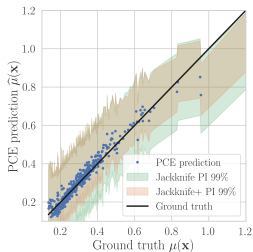
3D CFD model (ANSYS Icepak)

- $\#\mathcal{D} = 935$; $\mathcal{D}_{train} \leq 500$
- $\times 100$ data shuffling
- $P \in \{2, 3\}$, $s \in \{0.01, 0.05\}$

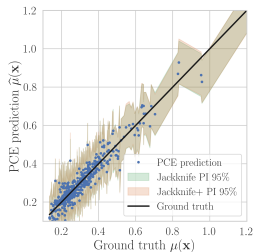
Parity plots



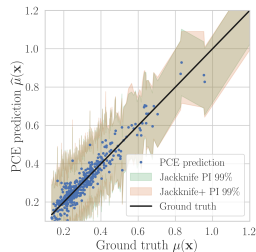
$P = 2, M = 250, s = 0.05$



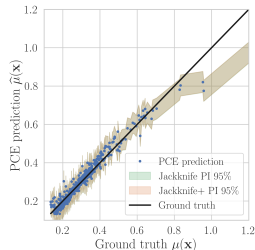
$P = 2, M = 250, s = 0.01$



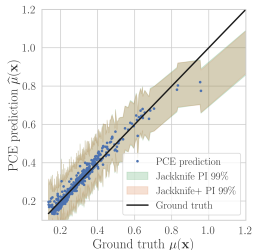
$P = 3, M = 250, s = 0.05$



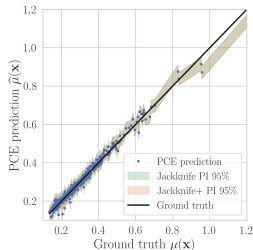
$P = 3, M = 250, s = 0.01$



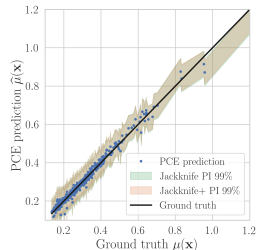
$P = 2, M = 500, s = 0.05$



$P = 2, M = 500, s = 0.01$

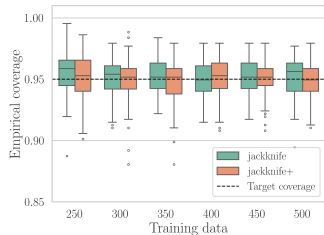


$P = 3, M = 500, s = 0.05$

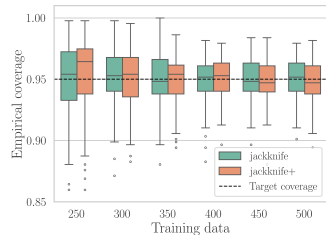


$P = 3, M = 500, s = 0.01$

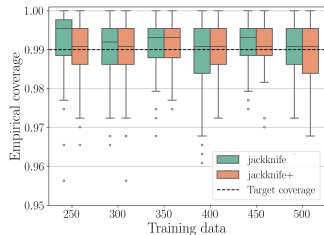
Coverage boxplots



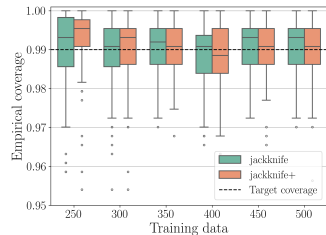
$P = 2, s = 0.05$



$P = 3, s = 0.05$

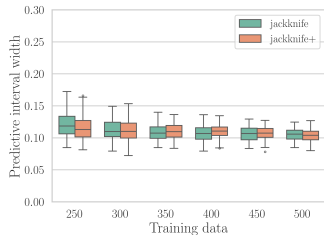


$P = 2, s = 0.01$

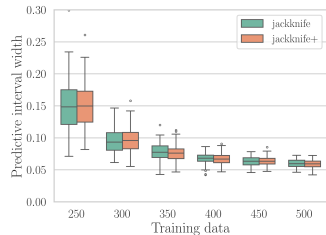


$P = 3, s = 0.01$

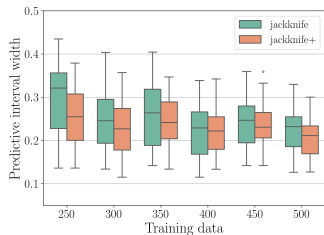
Interval width boxplots



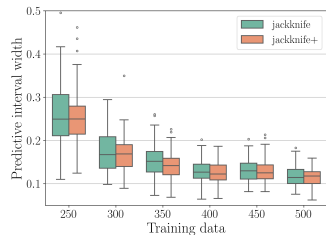
$P = 2, s = 0.05$



$P = 3, s = 0.05$



$P = 2, s = 0.01$



$P = 3, s = 0.01$

- 1 Surrogate modeling (data-driven)
- 2 Conformalized surrogate models
- 3 Engineering application
- 4 Summary & outlook

Take-away messages

- Conformal prediction complements point predictors with uncertainty intervals
- Little computational overhead for certain regression models
- Empirical coverage typically on par with target coverage
- Data availability is crucial for robustness

Ideas / questions

- ? Multi-output model outputs, e.g., field quantities
- ? Physics-informed/constrained CP
- ? Specifically for PCE: CP for sparse/adaptive PCEs